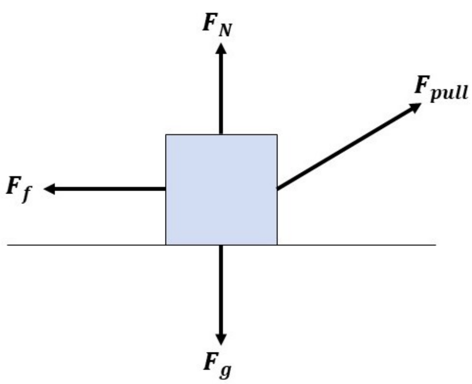
Quantum Algorithms for Cybersecurity, Chemistry, and Optimization

KEY IMAGES: Week 3

Hamiltonian Simulation > What is a Hamiltonian?

Newtonian Mechanics



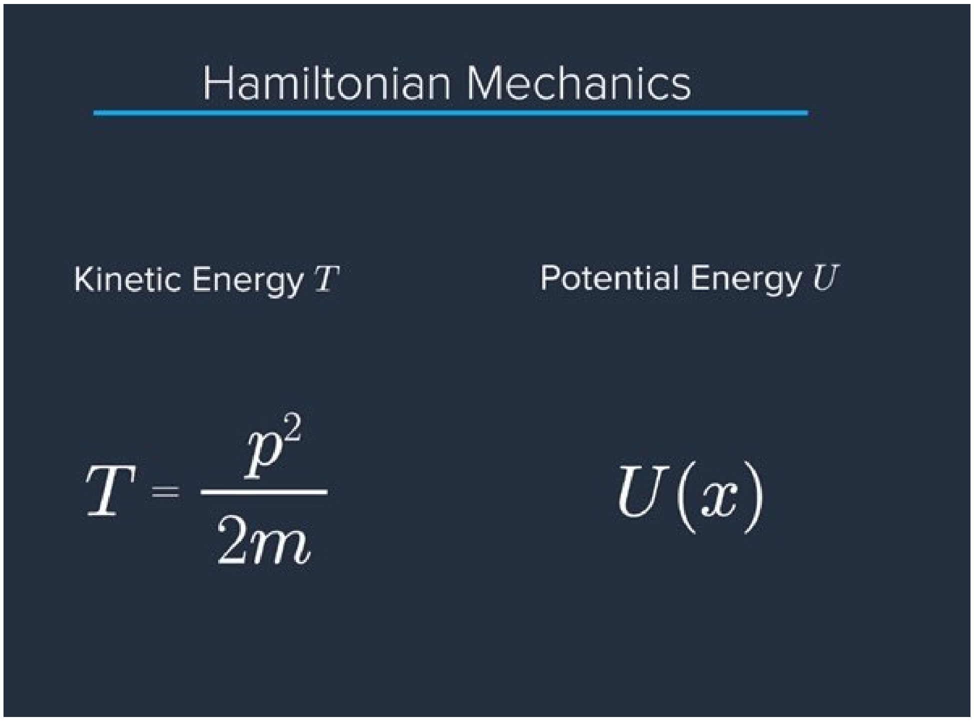
Newton's Second Law of Motion

$$\sum \boldsymbol{F} = \boldsymbol{ma}$$

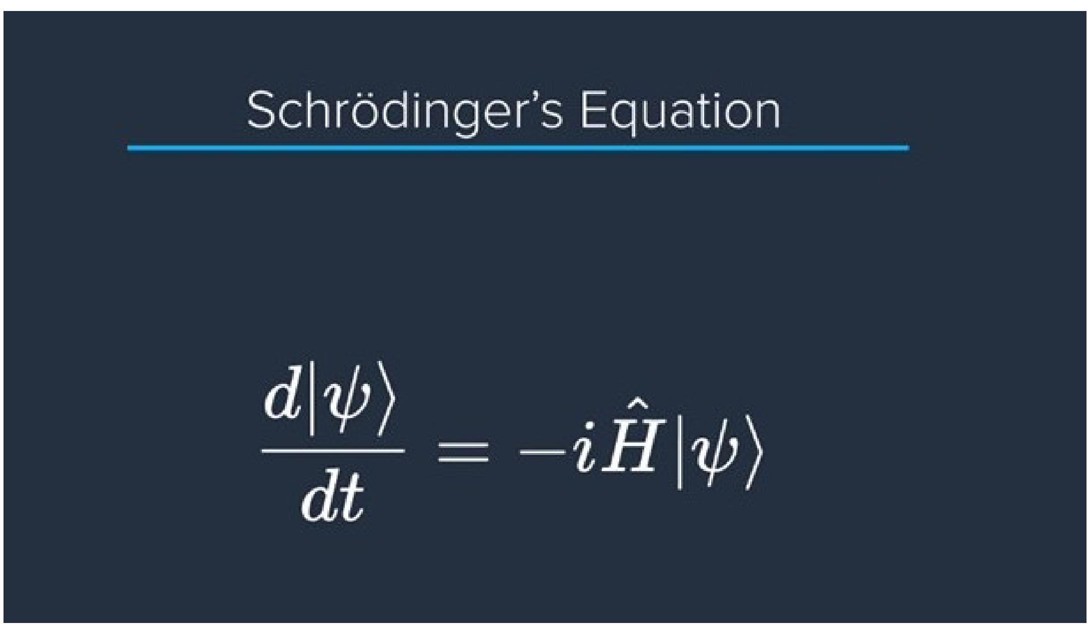
$$\sum \boldsymbol{F}(\boldsymbol{x}) = \boldsymbol{m}\frac{\boldsymbol{d^2x}}{\boldsymbol{dt^2}}$$

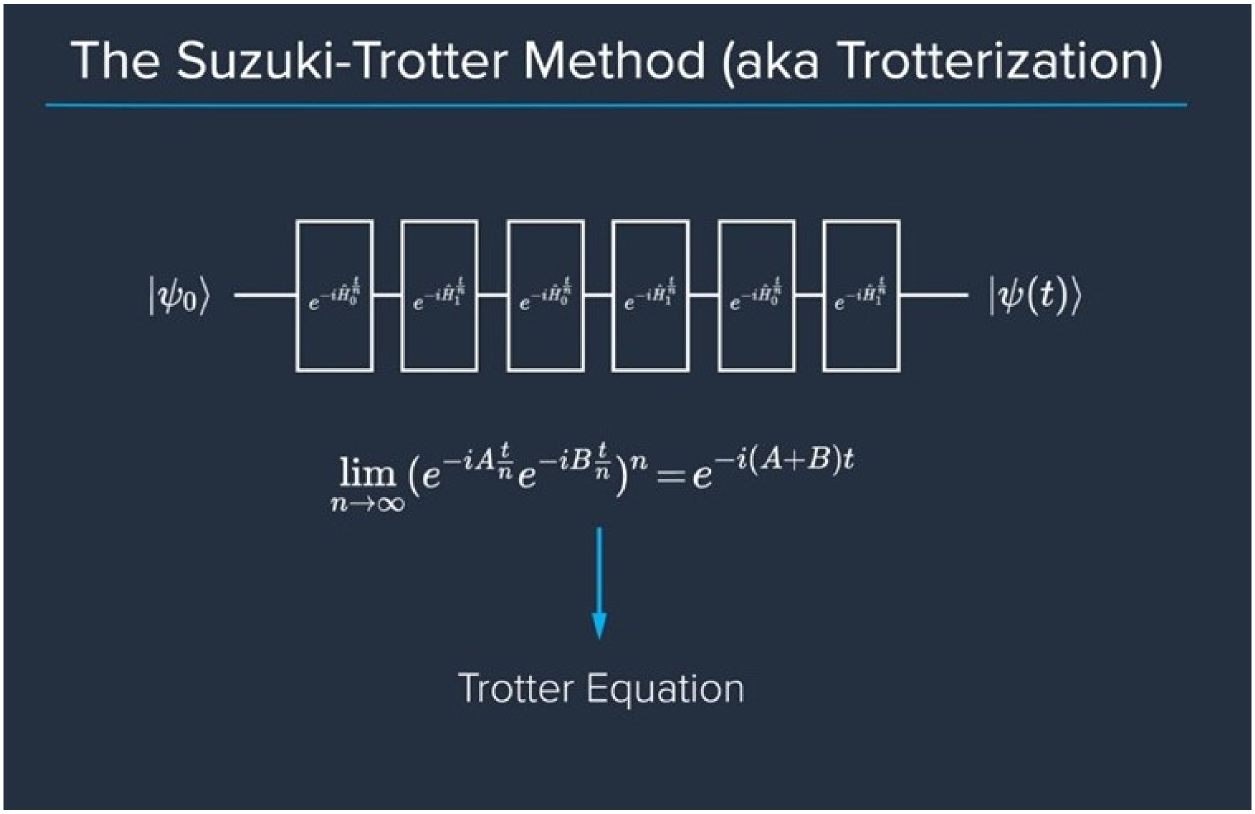
Solve this second order differential equation for x(t)

Hamiltonian Simulation > What is a Hamiltonian?



Hamiltonian Simulation > What is a Hamiltonian?




The Suzuki-Trotter Method (aka Trotterization)
$|\psi_0\rangle$
$e^{-i\hat{H}_0\frac{t}{n}}$
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$|\psi(t)\rangle$
$\lim_{n\to\infty}(e^{-iA\frac{t}{n}}e^{-iB\frac{t}{n}})^n = e^{-i(A+B)t}$
Trotter Equation

PHASE ESTIMATION